

43.7 Summary and Notes

43.7.1 Summary

Idea.

1. A local polynomial fit is weighted least squares solved afresh at every point; its weights are linear in $\mathbf{y}$ and reproduce polynomials of the fitted degree exactly, whatever the design and the bandwidth (Proposition 43.1.1).
2. Under a regular design, a bounded-variation kernel and a twice-differentiable f , the local linear estimate has bias $\frac{1}{2}\mu_2 f''(x)h^2 + o(h^2)$ and variance $\sigma^2 R(K)/\{nhg(x)\}$; the local constant estimate adds $\mu_2 h^2 f'g'/g$ in the interior and a term of order h at the boundary. The optimal bandwidth is of order $n^{-1/5}$ and the pointwise mean squared error of order $n^{-4/5}$ (Lemma 43.1.2, Theorem 43.1.3).
3. B-splines, defined by the de Boor recurrence, are nonnegative, supported on $d + 2$ consecutive knots, sum to one, are C^{d-1} at simple knots, have a derivative that is again a combination of B-splines, and form a basis of the spline space of Theorem 42.3.1 (Definition 43.2.1, Proposition 43.2.1); coefficients at the Greville abscissae reproduce the identity (Proposition 43.2.2). The basis matrix is banded and well conditioned where the truncated power basis is neither.
4. A penalized spline takes a rich B-spline basis and a difference penalty (Definition 43.3.1). The fit is a generalized ridge regression: it shrinks the Demmler–Reinsch coordinates by $(1 + \lambda e_j)^{-1}$, spends $\text{tr}(\mathbf{S}_\lambda) = \sum_j (1 + \lambda e_j)^{-1}$ degrees of freedom, and as $\lambda \rightarrow \infty$ tends to the least squares fit in the null space of the penalty — a constant for $k = 1$, a line for $k = 2$ (Lemma 43.3.1, Proposition 43.3.2, Proposition 43.3.3). Once the penalty is there, the number of knots stops mattering.
5. Among all functions with square-integrable second derivative, the minimizer of $\sum_i \{y_i - f(x_i)\}^2 + \lambda \int f''^2$ is the natural cubic spline with knots at the design points (Lemma 43.4.1, Theorem 43.4.2). Its fit solves a pentadiagonal system in $O(n)$ operations (Proposition 43.4.3).
6. Every smoother here is a linear operator $\mathbf{S}_\lambda$. Its trace measures what was spent, $\sigma^2 \mathbf{S}_\lambda \mathbf{S}_\lambda^\top$ is the covariance of the fit, and the obvious pointwise interval is an interval for $E\{\hat{f}(x_i)\}$, not for $f(x_i)$ (Proposition 43.5.1).
7. The smoothing parameter can be chosen by leave-one-out cross-validation, which needs no refitting, by generalized cross-validation, which needs only the trace, or by a corrected AIC: they agree in large samples with independent errors, differ in the upper tail, and all overfit badly when the errors are correlated (Theorem 43.5.2).
8. A quadratic penalty is a normal prior. Splitting the coefficients into the penalty's null space and its complement turns a penalized spline into a linear mixed model with $\lambda = \sigma^2/\sigma_u^2$, so that REML estimates the smoothing parameter and the posterior supplies a band (Theorem 43.6.1). That band is wider than the naive one by roughly the amount needed to absorb the bias on average — stated, not proved, in Proposition 43.6.2(c) — which buys across-the-function coverage and not pointwise coverage.

43.7.2 Notes and sources

Kernels and local polynomials. The local constant estimator is due independently to Nadaraya (1964) and Watson (1964), and the optimality of the kernel named after him to Epanechnikov (1969), though as Exercise (B2) shows the gain is a few per cent. The modern theory of local *polynomial* fitting is Fan and Gijbels (1996), whose chapter 3 has the random-design version of Theorem 43.1.3 with the conditional expansions handled properly; Wand and Jones (1995) give the background. The design-bias and boundary-bias comparisons in Theorem 43.1.3(b) and (c) are why local linear replaced Nadaraya–Watson, an argument made forcefully by Hastie and Loader (1993). The fixed-design proof given here, through Lemma 43.1.2, makes every error term explicit; the price is assumption (A), and Exercise (C2) says what changes without it. That the rate $n^{-4/5}$ cannot be improved for twice-differentiable f is Stone (1980, 1982), whose curse of dimensionality is what makes [Chapter 44](#) necessary. Loess is Cleveland (1979), extended by Cleveland and Devlin (1988), and plug-in bandwidths are Ruppert, Sheather and Wand (1995).

B-splines. Splines as a subject begin with Schoenberg (1946). The stable recurrence Eq. 43.2.2 is due to de Boor (1972) and Cox (1972); de Boor (2001) is the standard reference and contains Marsden’s identity, the Curry–Schoenberg theorem that the B-splines span the spline space for a general knot sequence, and the general form of Proposition 43.2.2. The proofs here need only the recurrence, the partition of unity and the derivative formula, and are stated for the clamped sequence; the induction in Proposition 43.2.1(f) is the shortest route I know to the basis property for it.

Penalized splines. Penalizing differences of a fitted sequence is Whittaker’s (1923) graduation, rediscovered many times; its marriage with the B-spline basis is Eilers and Marx (1996), which introduced both the name and the practice of taking many knots and letting the penalty do the work. O’Sullivan (1986) had earlier used the integral penalty on a B-spline basis, and Wand and Ormerod (2008) relate the two. Ruppert (2002) is the source of the rule of thumb for the knot count and Ruppert, Wand and Carroll (2003) the book-length treatment; Wood (2017) is the standard modern account of penalized regression splines and of the software that fits them, and [Chapter 44](#) draws on it. The simultaneous diagonalization of Lemma 43.3.1 goes back to Demmler and Reinsch (1975).

Smoothing splines. Reinsch (1967) gave the algorithm; the variational characterization Theorem 43.4.2 and the integration-by-parts argument behind it are due to Schoenberg and to Reinsch, and the treatment here follows Green and Silverman (1994, ch. 2), whose band matrices are those of Proposition 43.4.3. Wahba (1990) is the definitive account from the reproducing-kernel side, with the representer theorem of Kimeldorf and Wahba (1971) at its centre; Eubank (1999) is gentler. Silverman (1984) computed the equivalent kernel quoted in [Section 43.4](#), and Silverman (1985) surveys the approach; the $O(n)$ computation of the diagonal of $\mathbf{S}_\lambda$ is Hutchinson and de Hoog (1985).

Choosing the smoothing parameter. Generalized cross-validation is Craven and Wahba (1979), and the leave-one-out shortcut for quadratic penalties is Theorem 29.3.2. The corrected AIC Eq. 43.5.5 is Hurvich, Simonoff and Tsai (1998), who make the case that its pole is a feature. Opsomer, Wang and Yang (2001) review what correlated errors do, which is what Example (What correlated errors do) illustrates: no automatic rule separates a smooth mean from smooth noise without an assumption. Reiss and Ogden (2009) compare REML and GCV. Nothing here addresses inference *after* the selection except by saying so; Proposition 29.5.1 is the warning, and the remedies are those of [Chapter 23](#) and [Chapter 29](#).

Mixed models and Bayes. The equivalence between a smoothing penalty and a random

effect is old — implicit in Whittaker’s graduation and explicit in Wahba (1978) — and was made into a working method by Ruppert, Wand and Carroll (2003), whose chapter 4 contains the transformation of Theorem 43.6.1(a) in essentially the form given here. Wahba (1983) introduced the Bayesian band and Nychka (1988) analysed its coverage, coining the across-the-function reading that Proposition 43.6.2(c) states; that part is quoted, not proved, and the simulation in Example (Measuring the coverage) is the evidence offered for it. Sun and Loader (1994) give genuinely simultaneous bands. The adaptive methods named at the end of [Section 43.6](#) are Denison, Mallick and Smith (1998), Donoho and Johnstone (1994) and Ramsay (1988).

Coverage. This chapter follows the ground of Fahrmeir, Kneib, Lang and Marx (2021, §8.1 and §8.3); their §8.2, on bivariate and spatial smoothing, belongs to [Chapter 44](#), as does Hastie and Tibshirani (1990), the book that made smoothers part of the regression curriculum.

43.7.3 References

- Cleveland, William S. (1979). Robust Locally Weighted Regression and Smoothing Scatterplots. *Journal of the American Statistical Association* 74(368), 829–836.
- Cleveland, William S. and Devlin, Susan J. (1988). Locally Weighted Regression: An Approach to Regression Analysis by Local Fitting. *Journal of the American Statistical Association* 83(403), 596–610.
- Cox, Maurice G. (1972). The Numerical Evaluation of B-Splines. *Journal of the Institute of Mathematics and Its Applications* 10(2), 134–149.
- Craven, Peter and Wahba, Grace (1979). Smoothing Noisy Data with Spline Functions. *Numerische Mathematik* 31(4), 377–403.
- de Boor, Carl (1972). On Calculating with B-Splines. *Journal of Approximation Theory* 6(1), 50–62.
- de Boor, Carl (2001). *A Practical Guide to Splines*. Revised edition. New York: Springer.
- Demmler, A. and Reinsch, Christian (1975). Oscillation Matrices with Spline Smoothing. *Numerische Mathematik* 24(5), 375–382.
- Denison, David G. T., Mallick, Bani K. and Smith, Adrian F. M. (1998). Automatic Bayesian Curve Fitting. *Journal of the Royal Statistical Society, Series B* 60(2), 333–350.
- Donoho, David L. and Johnstone, Iain M. (1994). Ideal Spatial Adaptation by Wavelet Shrinkage. *Biometrika* 81(3), 425–455.
- Eilers, Paul H. C. and Marx, Brian D. (1996). Flexible Smoothing with B-splines and Penalties. *Statistical Science* 11(2), 89–121.
- Epanechnikov, V. A. (1969). Non-parametric Estimation of a Multivariate Probability Density. *Theory of Probability and Its Applications* 14(1), 153–158.
- Eubank, Randall L. (1999). *Nonparametric Regression and Spline Smoothing*. 2nd edition. New York: Marcel Dekker.
- Fahrmeir, Ludwig, Kneib, Thomas, Lang, Stefan and Marx, Brian D. (2021). *Regression: Models, Methods and Applications*. 2nd edition. Berlin: Springer.
- Fan, Jianqing and Gijbels, Irène (1996). *Local Polynomial Modelling and Its Applications*. London: Chapman and Hall.
- Green, Peter J. and Silverman, Bernard W. (1994). *Nonparametric Regression and Generalized Linear Models: A Roughness Penalty Approach*. London: Chapman and Hall.
- Hastie, Trevor and Loader, Clive (1993). Local Regression: Automatic Kernel Carpentry. *Statistical Science* 8(2), 120–129.
- Hastie, Trevor and Tibshirani, Robert (1990). *Generalized Additive Models*. London: Chapman and Hall.
- Hurvich, Clifford M., Simonoff, Jeffrey S. and Tsai, Chih-Ling (1998). Smoothing Parameter Selection in Nonparametric Regression Using an Improved Akaike Information Criterion. *Journal of the Royal Statistical Society, Series B* 60(2), 271–293.

- Hutchinson, Michael F. and de Hoog, Frank R. (1985). Smoothing Noisy Data with Spline Functions. *Numerische Mathematik* 47(1), 99–106.
- Kimeldorf, George and Wahba, Grace (1971). Some Results on Tchebycheffian Spline Functions. *Journal of Mathematical Analysis and Applications* 33(1), 82–95.
- Nadaraya, Elizbar A. (1964). On Estimating Regression. *Theory of Probability and Its Applications* 9(1), 141–142.
- Nychka, Douglas (1988). Bayesian Confidence Intervals for Smoothing Splines. *Journal of the American Statistical Association* 83(404), 1134–1143.
- Opsomer, Jean, Wang, Yuedong and Yang, Yuhong (2001). Nonparametric Regression with Correlated Errors. *Statistical Science* 16(2), 134–153.
- O’Sullivan, Finbarr (1986). A Statistical Perspective on Ill-posed Inverse Problems. *Statistical Science* 1(4), 502–518.
- Ramsay, James O. (1988). Monotone Regression Splines in Action. *Statistical Science* 3(4), 425–441.
- Reinsch, Christian H. (1967). Smoothing by Spline Functions. *Numerische Mathematik* 10(3), 177–183.
- Reiss, Philip T. and Ogden, R. Todd (2009). Smoothing Parameter Selection for a Class of Semiparametric Linear Models. *Journal of the Royal Statistical Society, Series B* 71(2), 505–523.
- Ruppert, David (2002). Selecting the Number of Knots for Penalized Splines. *Journal of Computational and Graphical Statistics* 11(4), 735–757.
- Ruppert, David, Sheather, Simon J. and Wand, Matthew P. (1995). An Effective Bandwidth Selector for Local Least Squares Regression. *Journal of the American Statistical Association* 90(432), 1257–1270.
- Ruppert, David, Wand, Matthew P. and Carroll, Raymond J. (2003). *Semiparametric Regression*. Cambridge: Cambridge University Press.
- Schoenberg, Isaac J. (1946). Contributions to the Problem of Approximation of Equidistant Data by Analytic Functions. *Quarterly of Applied Mathematics* 4, 45–99 and 112–141.
- Silverman, Bernard W. (1984). Spline Smoothing: The Equivalent Variable Kernel Method. *The Annals of Statistics* 12(3), 898–916.
- Silverman, Bernard W. (1985). Some Aspects of the Spline Smoothing Approach to Nonparametric Regression Curve Fitting. *Journal of the Royal Statistical Society, Series B* 47(1), 1–52.
- Stone, Charles J. (1980). Optimal Rates of Convergence for Nonparametric Estimators. *The Annals of Statistics* 8(6), 1348–1360.
- Stone, Charles J. (1982). Optimal Global Rates of Convergence for Nonparametric Regression. *The Annals of Statistics* 10(4), 1040–1053.
- Sun, Jiayang and Loader, Clive R. (1994). Simultaneous Confidence Bands for Linear Regression and Smoothing. *The Annals of Statistics* 22(3), 1328–1345.
- Wahba, Grace (1978). Improper Priors, Spline Smoothing and the Problem of Guarding Against Model Errors in Regression. *Journal of the Royal Statistical Society, Series B* 40(3), 364–372.
- Wahba, Grace (1983). Bayesian “Confidence Intervals” for the Cross-Validated Smoothing Spline. *Journal of the Royal Statistical Society, Series B* 45(1), 133–150.
- Wahba, Grace (1990). *Spline Models for Observational Data*. Philadelphia: SIAM.
- Wand, Matthew P. and Jones, M. Chris (1995). *Kernel Smoothing*. London: Chapman and Hall.
- Wand, Matthew P. and Ormerod, John T. (2008). On Semiparametric Regression with O’Sullivan Penalized Splines. *Australian and New Zealand Journal of Statistics* 50(2), 179–198.
- Watson, Geoffrey S. (1964). Smooth Regression Analysis. *Sankhyā: The Indian Journal*

- of Statistics, Series A* 26(4), 359–372.
- Whittaker, Edmund T. (1923). On a New Method of Graduation. *Proceedings of the Edinburgh Mathematical Society* 41, 63–75.
 - Wood, Simon N. (2017). *Generalized Additive Models: An Introduction with R*. 2nd edition. Boca Raton: Chapman and Hall/CRC.